

ID:

Name:

Section:

BRAC University

Course Code: CSE350

Digital Electronics and Pulse Techniques

Faculty: QSH

Set

A

Assessment: Quiz 2

Duration: 20 minutes

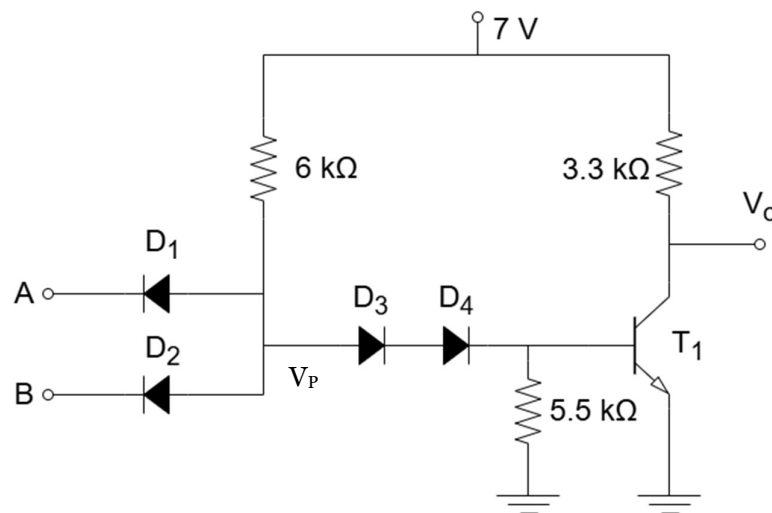
Full Marks: 20



Use $V_f(\text{diode}) = 0.6\text{ V}$, $V_D(\text{conduction voltage of diode}) = 0.7\text{ V}$, $V_f(\text{transistor}) = 0.5\text{ V}$, $V_{BE}(\text{forward active}) = 0.7\text{ V}$, $V_{BE}(\text{saturation}) = 0.8\text{ V}$, $V_{CE}(\text{saturation}) = 0.2\text{ V}$, $\beta_F = 30$, $V_{OH} = 5\text{ V}$ for all the questions.

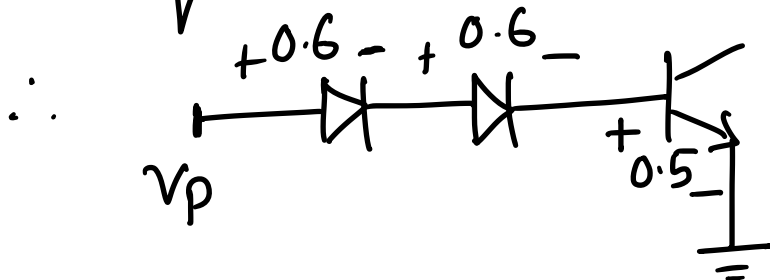
(a) [20 marks]

- [1] Which logic gate does this circuit represent?
- [2] Determine the **Minimum Voltage** required at V_P to turn ON D_3 , D_4 and T_1
- [3] Determine the **Maximum Fanout**
- [4] Determine the **Average Power Dissipation**



(1) NAND Gate

(2) Use cut in voltage for Minimum voltage required.



$$\begin{aligned}
 \text{Minimum } V_P &= 0.6 + 0.6 + 0.5 \\
 &= 1.7\text{ V}
 \end{aligned}$$

(3) Case 1: Output High $\rightarrow$ Fanout $= \infty$

Case 2: Output Low, $V_O = 0.2 \text{ V}$

$$\therefore N < \frac{\beta_F I_B - I_{C'}}{I_L}$$

$$N < \frac{30 \times 0.655 - 2.06}{1.016}$$

$$\therefore N < 17.313$$

$$\therefore N = 17$$

$$I_{C'} = \frac{7 - 0.2}{3.3k} = 2.06 \text{ mA}$$

$$I_B = I_1 - I_2$$

$$= \frac{7 - 2.2}{6k} - \frac{0.8 - 0}{5.5k}$$

$$= 0.8 - 0.145$$

$$= 0.655 \text{ mA}$$

$$I_L = \frac{7 - 0.9}{6k} = 1.016 \text{ mA}$$

Maximum Fanout $= \min(\infty, 17) = 17$

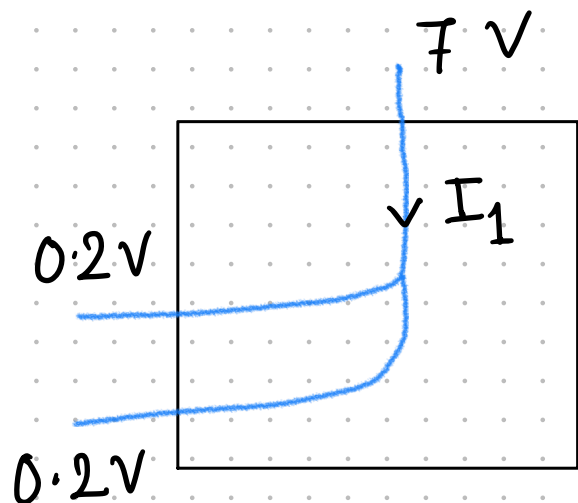
(4) Power dissipation

Case 1, 2, 3

$$P = \Delta V I$$

$$= (7 - 0.2) \times \frac{7 - 0.9}{6k}$$

$$= 6.913 \text{ mW}$$

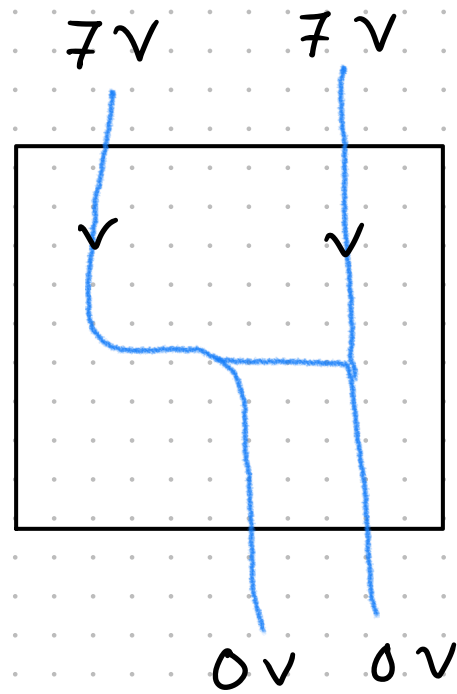


Case 4

$$P = \sum \Delta V I$$

$$= (7-0) \left(\frac{7-2.2}{6k} + \frac{7-0.2}{3.3k} \right)$$

$$= 20.024 \text{ mW}$$



$$\text{Average Power Dissipation} = \frac{3 \times 6.913 + 20.024}{4}$$

$$= 10.191 \text{ mW}$$